CS430/910: Foundations of Data Analytics

Clustering | Dr Ayse Sunar

Part A: Introduction to Clustering

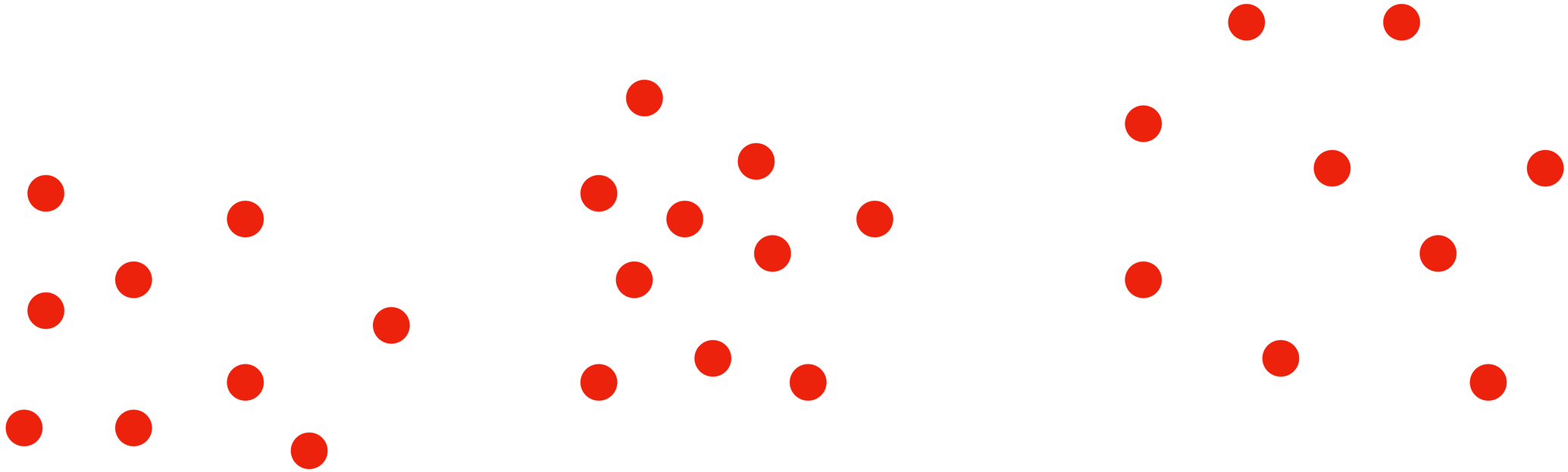
Objectives

- To understand the concept of clustering and clustering methods
- To see different clustering methods and objectives:
  - k-**means**
  - k-**medoids**
  - **Hierarchical Agglomerative Clustering**
  - **DBSCAN**
- To use tools (Weka) to cluster data and study the results

Clustering

- **Classification** learns classes based on **labeled** training data (supervised learning)
- **Clustering** aims to find classes **without labeled** examples
  - An "**unsupervised**" learning method
  - Place similar items in same group, different items in different groups

What is Clustering?



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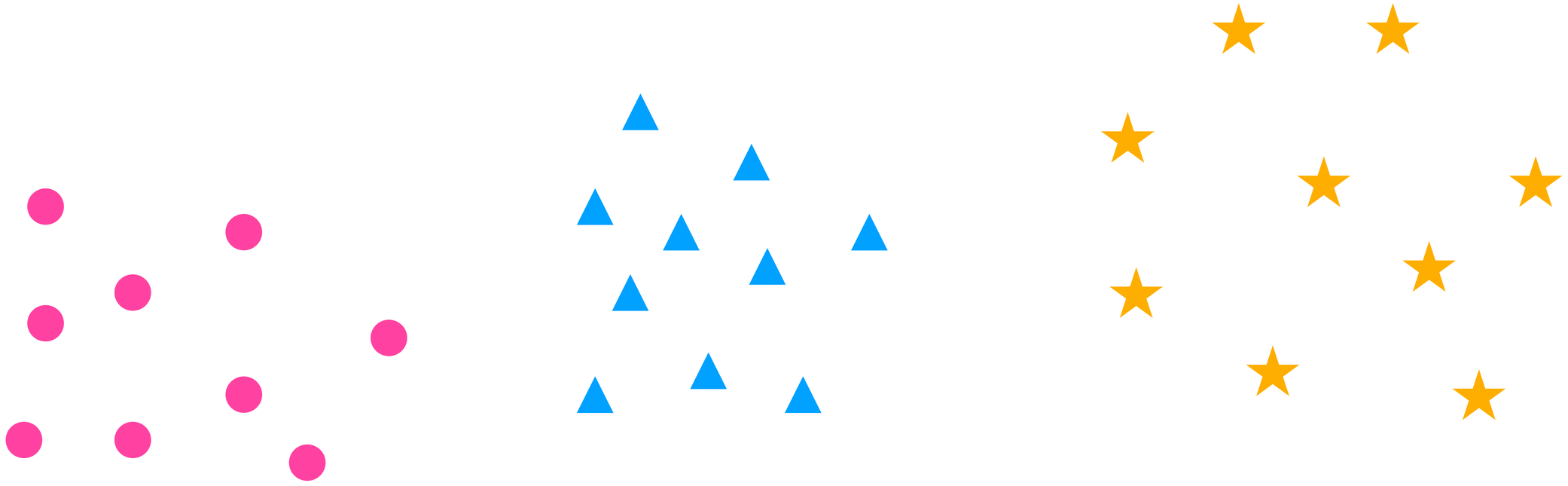
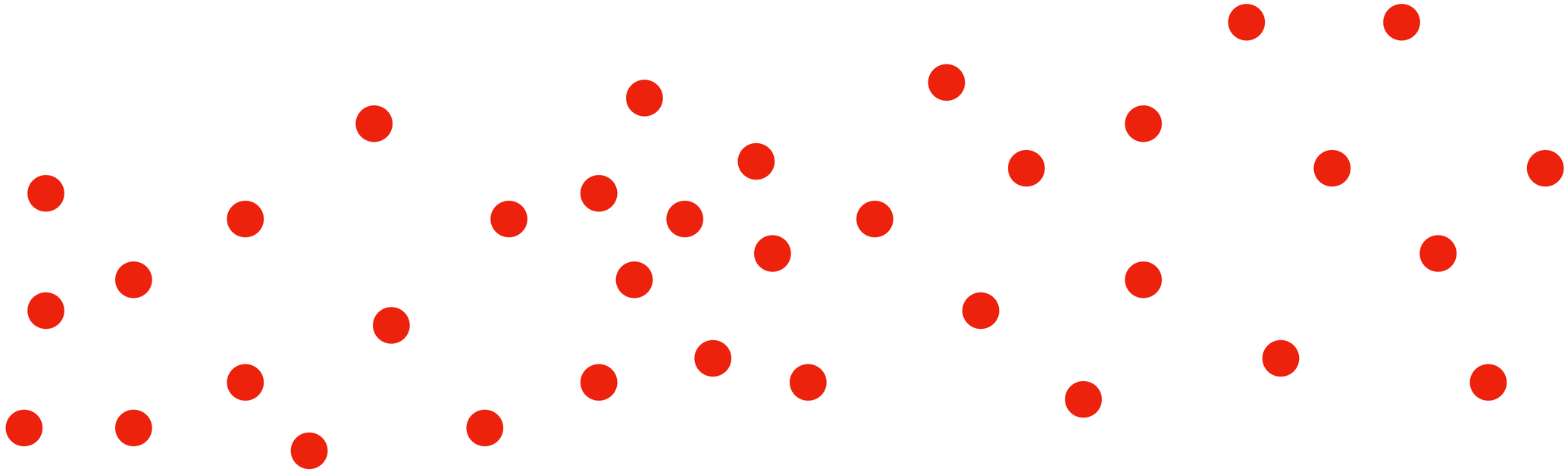


Clustering

It’s Not Always Easy…



Clustering

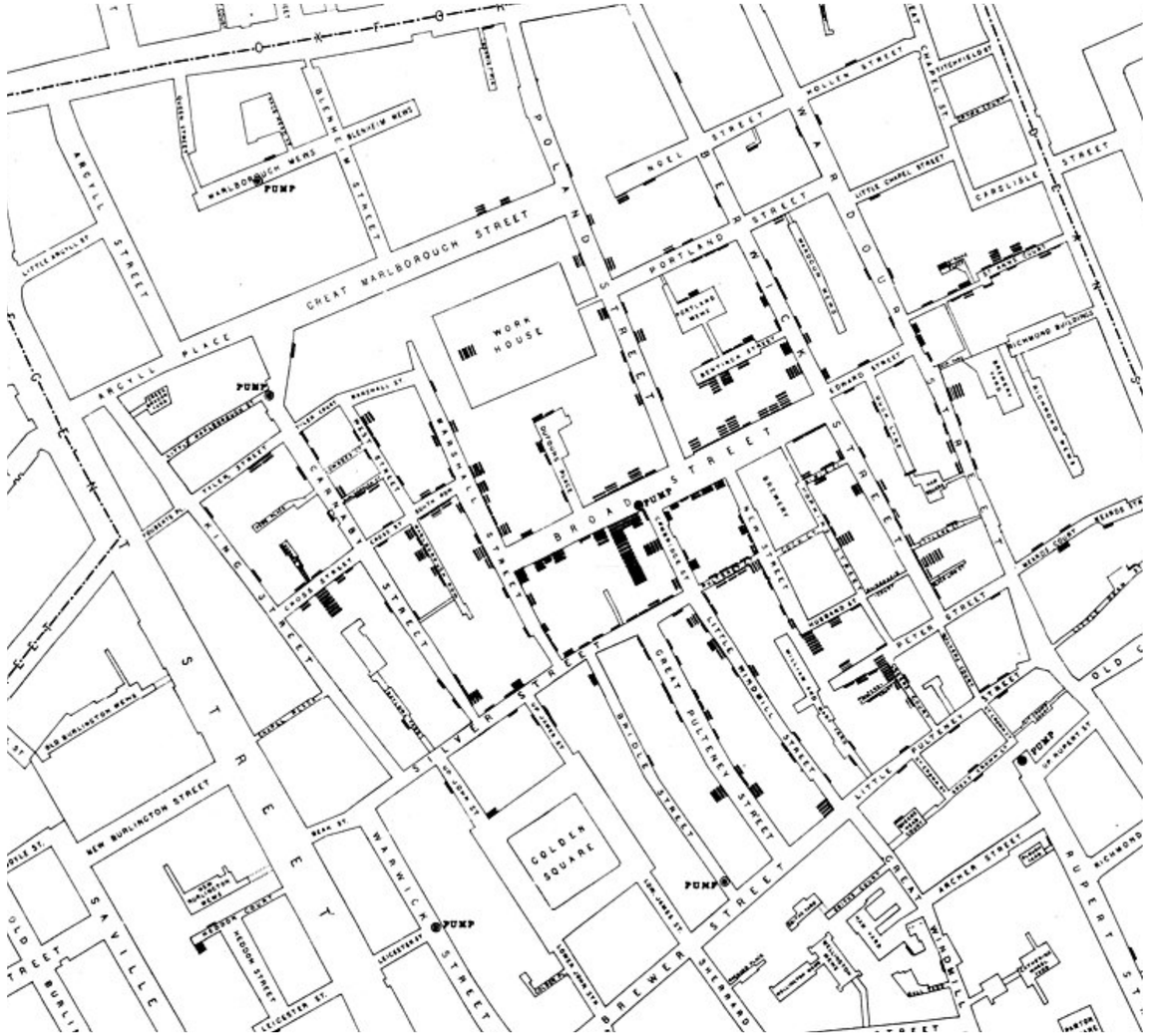
- Many different uses for clustering:
  - **Understand** data better: find groups in data
  - **Identify possible classes** that can be predicted by classification
  - Extract **representative examples** from each cluster
  - **Data Reduction**: only work with a few examples per cluster
  - As a **preprocessing** step: identify outliers
  - To **split up** data: process each cluster on a different machine
  - To **speed up similarity search**: only look within certain clusters

Applications

- **Biology/Genetics**: identify similar entities (organisms/genomes)
- **Information Retrieval**: find clusters of similar documents
- **Marketing**: identify customers with similar behaviour
- **Urban Planning**: organise regions into similar land-use
- **Climate**: find patterns of weather behaviour
- **Sociology**: find groups of people with similar views

An Early Application

- John Snow plotted the location of cholera cases on a map during an outbreak in the summer of 1854
- His hypothesis was that the disease was carried in water, so he plotted location of cases and water pumps, identifying the source
- Clusters easy to identify in two dimensions, more points and higher dimension?



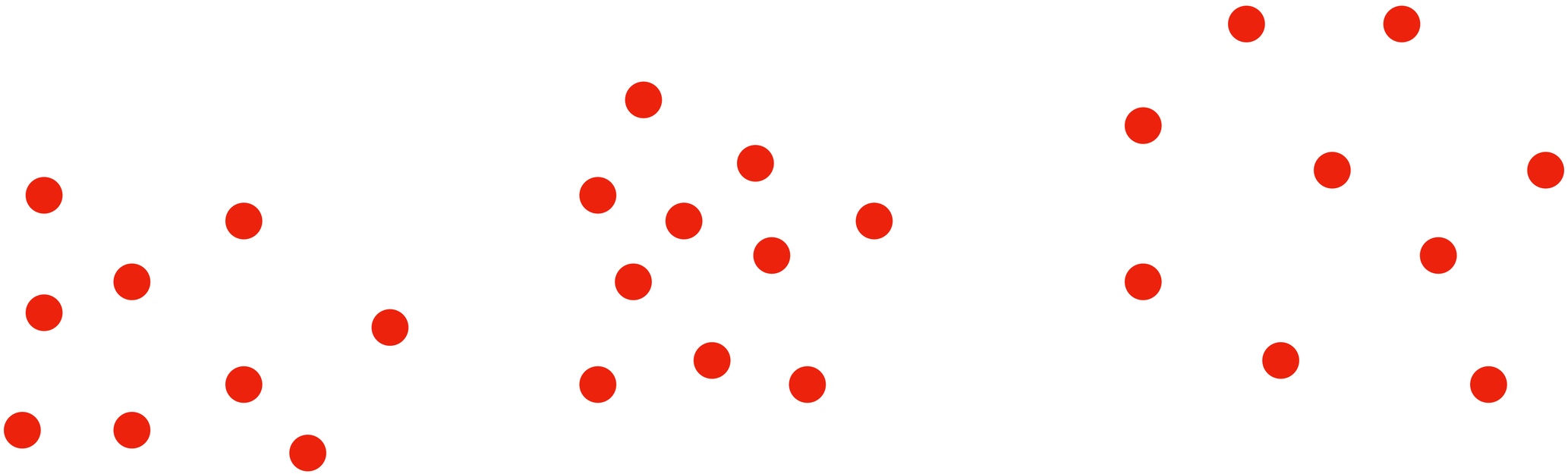
Snow, J., 1856. On the mode of communication of cholera. *Edinburgh medical journal*, *1*(7), p.668.
Ph.ucla.edu. n.d. *High resolution maps of John Snow*. [online] Available at: <https://www.ph.ucla.edu/epi/snow/highressnowmap.html> [Accessed 30 July 2021].

Overview

- Clustering has an intuitive appeal
  - People often talk “informally” about “clusters”: “cancer clusters”, “disease clusters”, “crime clusters”, …
- Will define what is meant by clustering, formalise the goals of clustering, and give algorithms for clustering data
  - Which **attributes** will be used for clustering, what is an appropriate **distance** between objects to use, what **objective** should the clustering optimise, **how many** clusters should we aim to find, how to **evaluate** the results?

Clustering

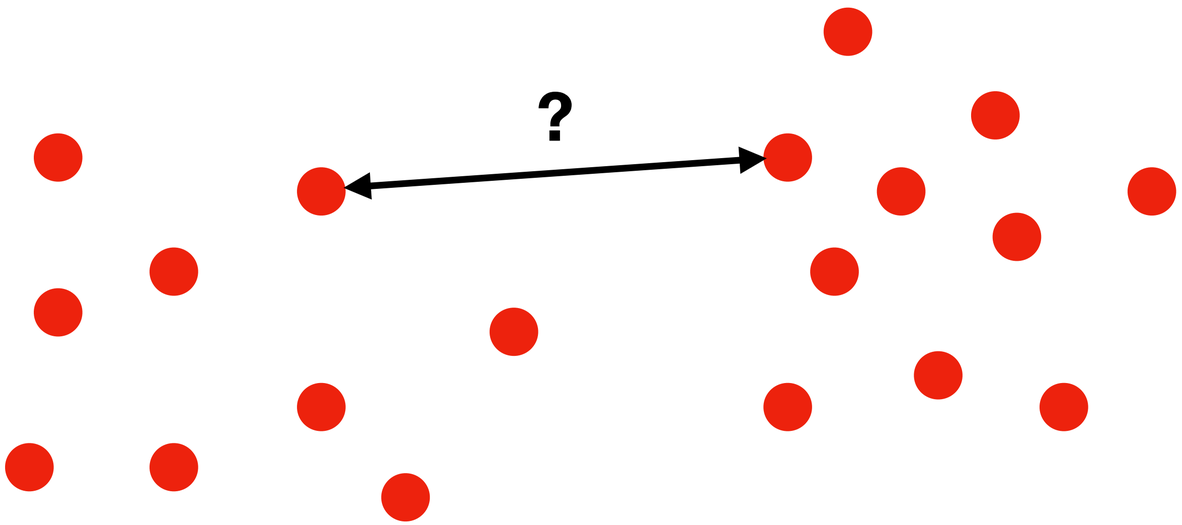
Objectives

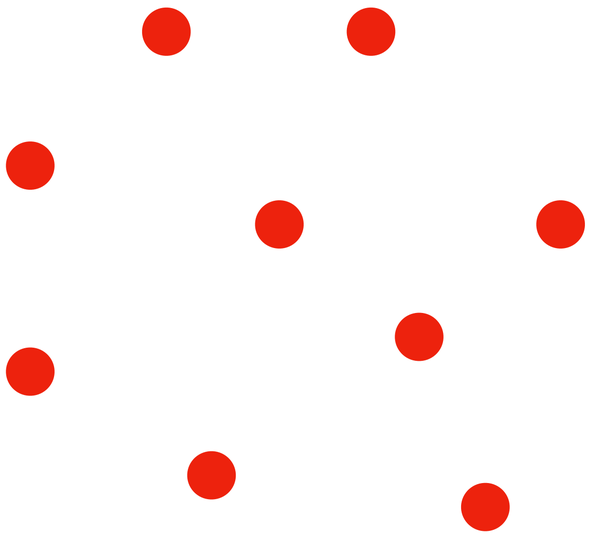


Informally, there are items, and we want to **group** them into **clusters.**

Clustering

Objectives



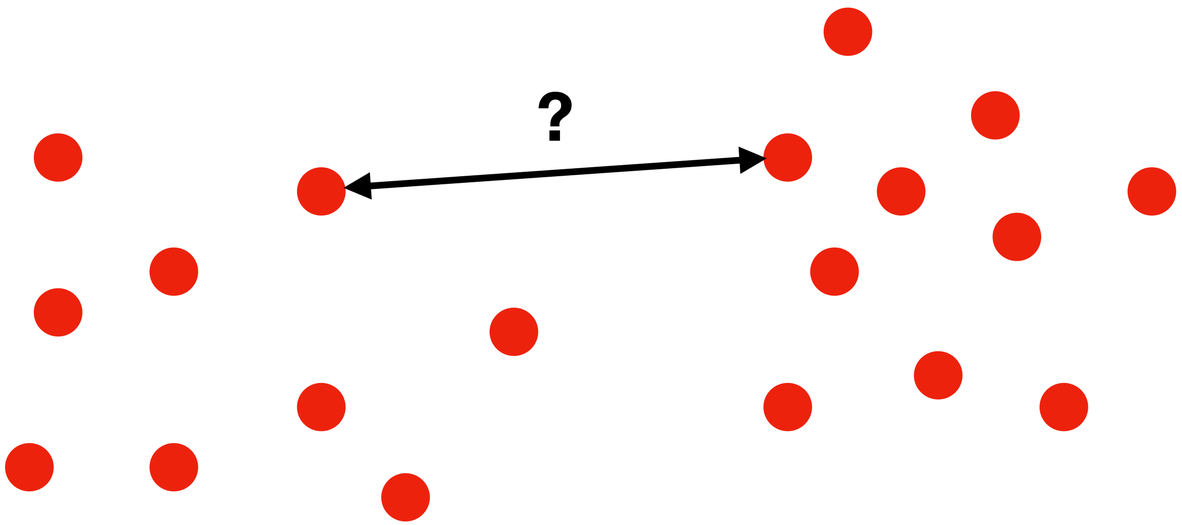


But how do we measure the distance between points?

Informally, there are items, and we want to **group** them into **clusters.**

Clustering

Objectives



But how do we measure the distance between points?

It's obvious,… right? Maybe in 2D… or is it?

We could use the Euclidean distance.

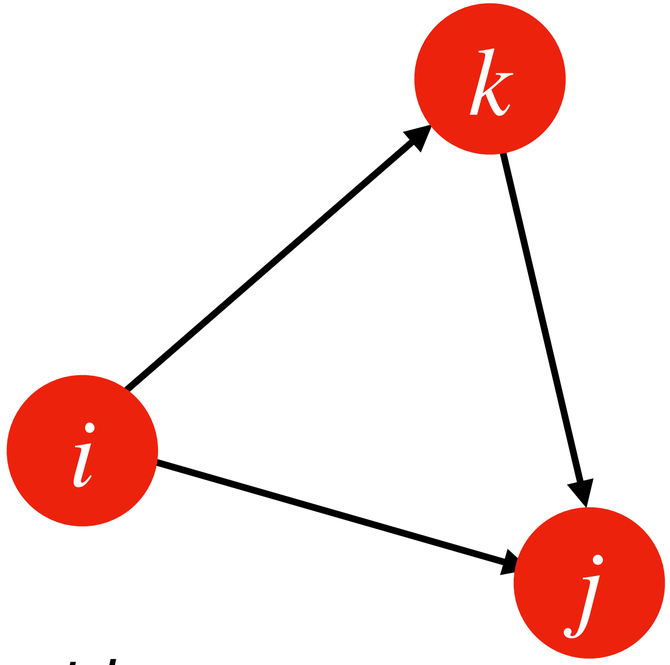
But what about data that is a mix of different types (e.g. time, text, boolean values). Should we weight different attributes?

Some clustering algorithms assume a certain distance function, with many assuming Euclidean distance.

Distances Refresher

- The most common distance measures satisfy the following mathematical properties:

  1. **Non-negativity**: $d(i, j) \geq 0$
  2. **Identity**: $d(i, i) = 0$
  3. **Symmetry**: $d(i, j) = d(j, i)$
  4. **Triangle Inequality**: $d(i, j) \leq d(i, k) + d(k, j)$



- Properties which obey these conditions are known as *metrics*.

Distances

- Most distance measurements of interest are metrics:
  - Euclidean distance ($d(i,j) = \sqrt{(i_1 - j_1)^2 + (i_2 - j_2)^2 + \ldots + (i_n - j_n)^2}$)
  - L_1 distance ($d(i,j) = \sum_{k=1}^{n} |i_k - j_k|$, sum of absolute coordinate differences)
  - L_∞ distance ($d(i,j) = max_k(|i_k - j_k|)$, maximum absolute coordinate difference)
  - String Edit distance (number of changes to turn one string into another):
    - The Hamming distance is a type of string edit distance.

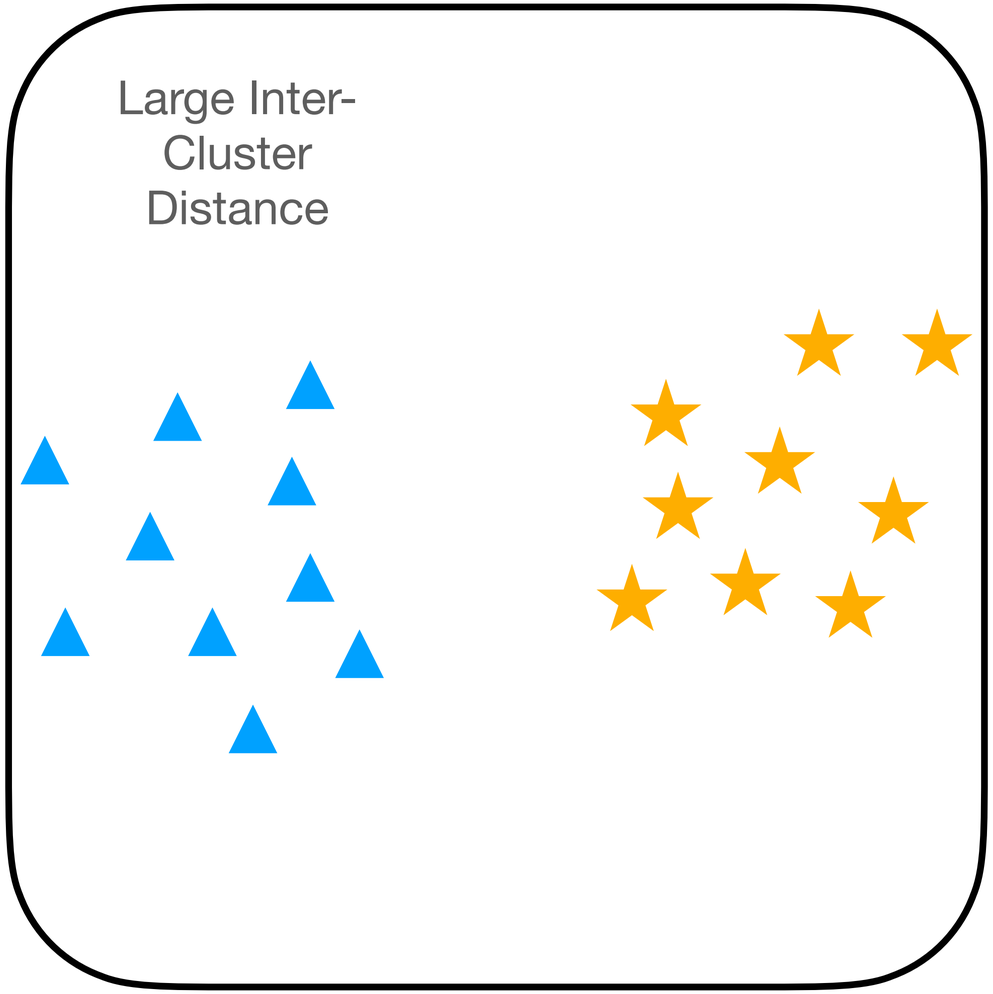
Clustering

Objectives

- Divide data into k clusters C so that
  - Inter-cluster distance (between clusters) is large
    - The clusters are **well-separated**
  - Intra-cluster distance (within cluster) is small
    - The clusters are **coherent**

Clustering

Inter- vs. Intra-Cluster Distance



Clustering

Objectives

- Divide data into k clusters C so that
  - Inter-cluster distance (between clusters) is large
    - The clusters are **well-separated**
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    - The clusters are **coherent**
- Formalise this into a mathematical objective.
  - Eg. Formalise some measure of **Inter** distance $D(C)$,
  - Formalise some measure of **Intra** distance $T(C)$,
  - Try to maximise $(D(C) - T(C))$

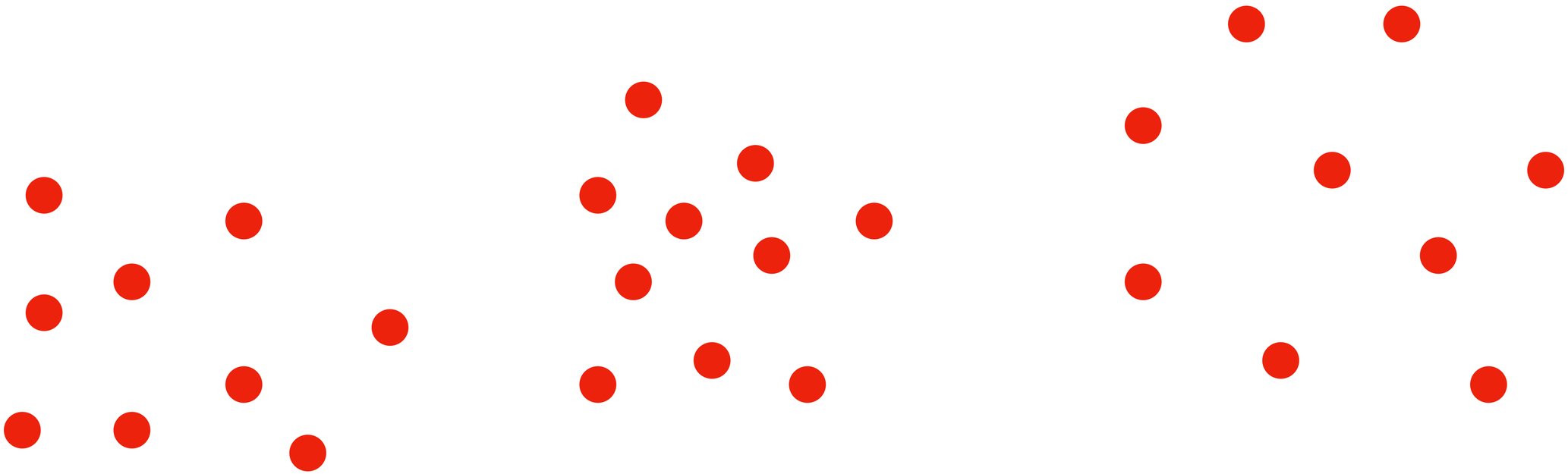
Part B: Clustering Algorithms

k-Means

1. Specify k (how many clusters you want).
2. Choose k objects (or points) at random from the data as cluster centres.
3. All data objects are assigned to their closest cluster centre (using squared Euclidean distance).
4. The mean (or *centroid*) of each cluster is calculated. These become the new cluster centres.
5. Repeat steps 3-4 until convergence.

k-Means

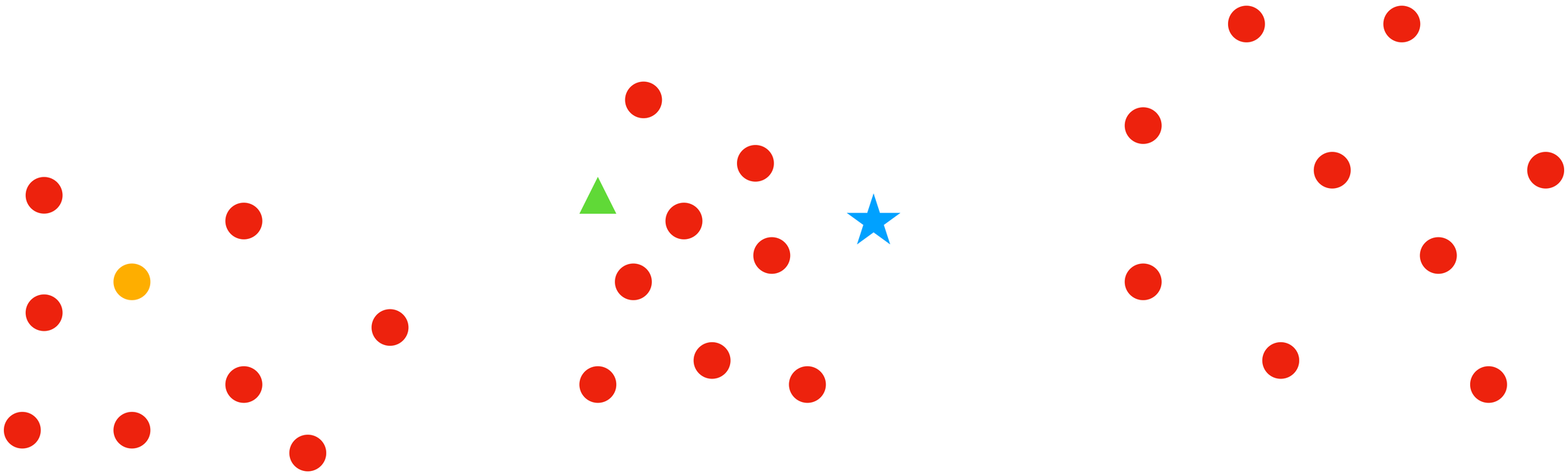
Example



Start with this data.

k-Means

Example

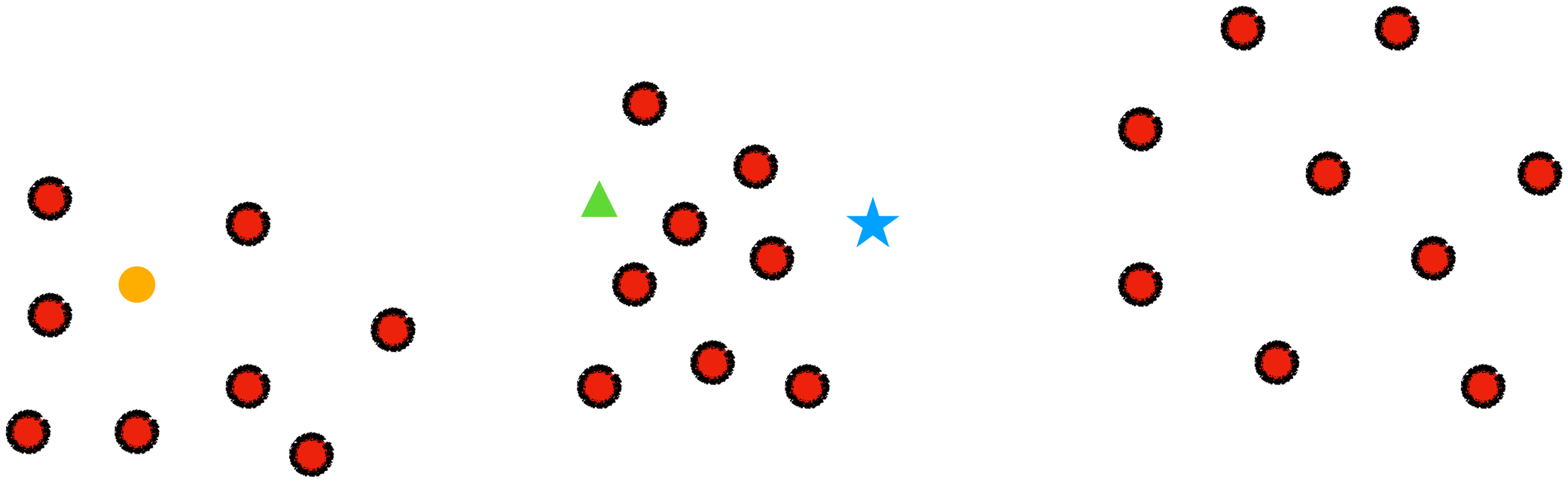


1. Specify k (lets say 3).

2. Choose k objects at random from the data as cluster centres.

k-Means

Example



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k-Means

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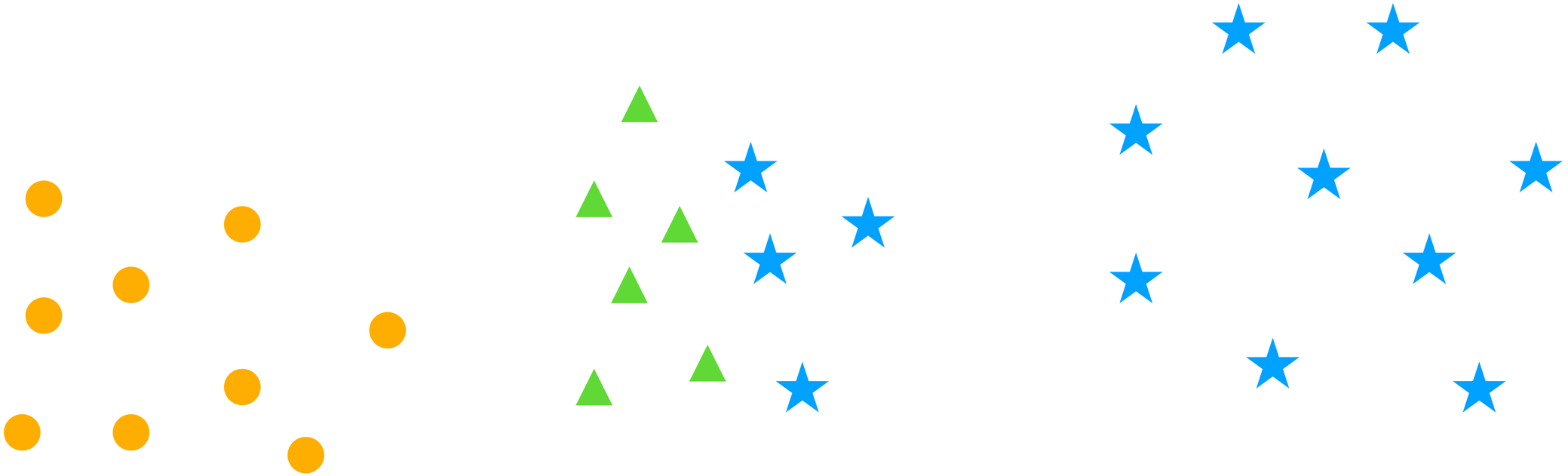

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k-Means

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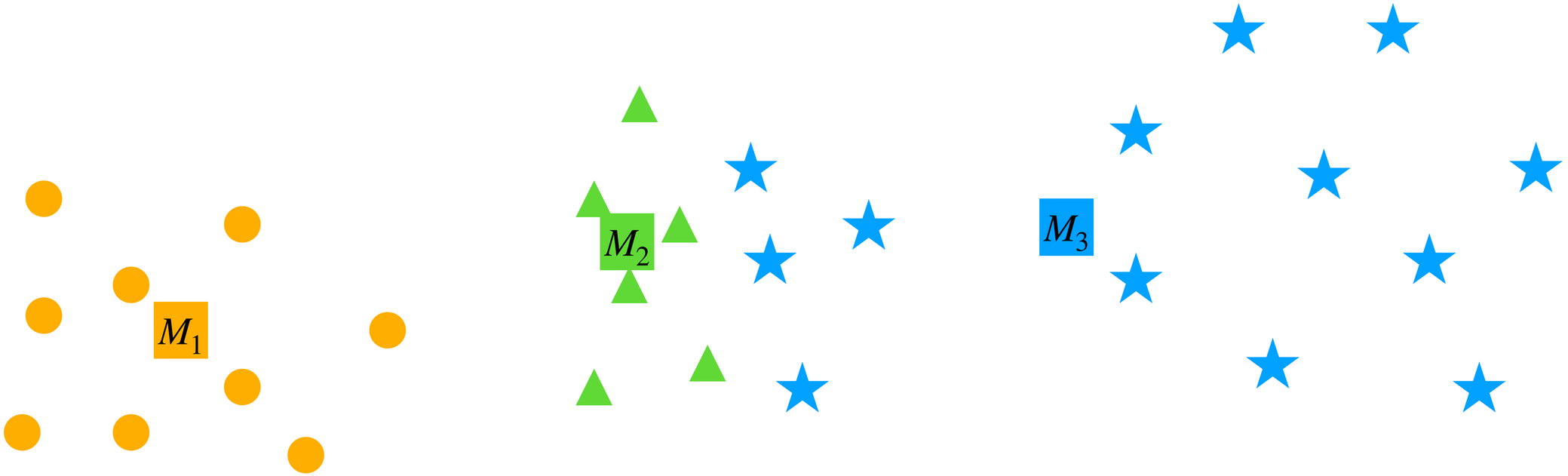

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k-Means

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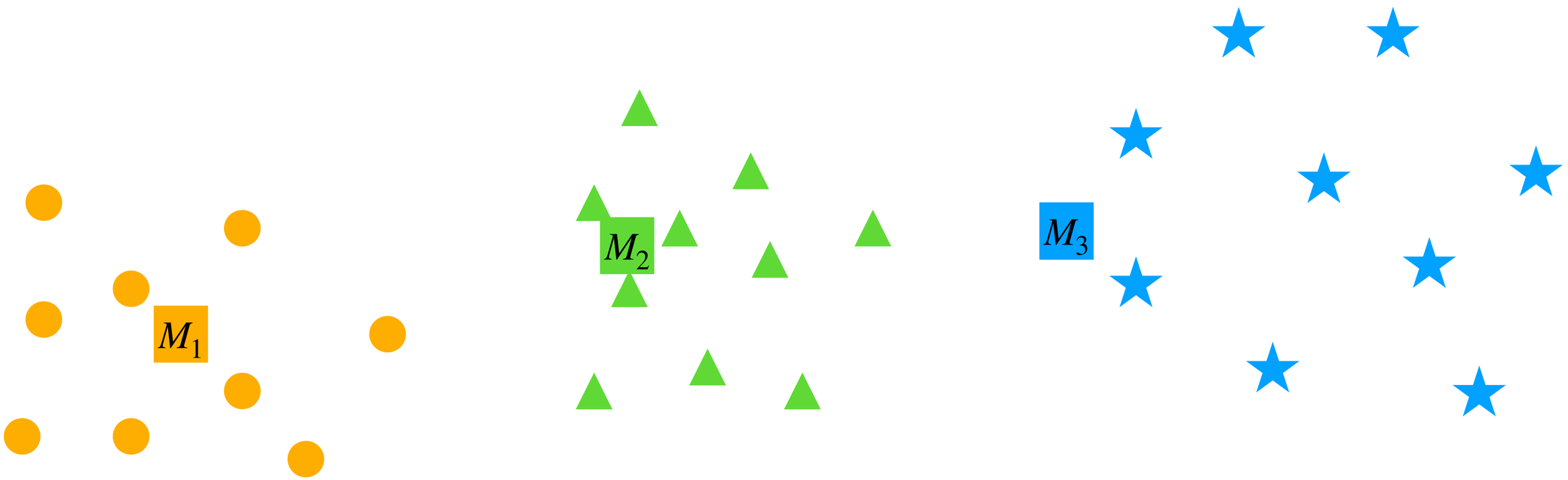

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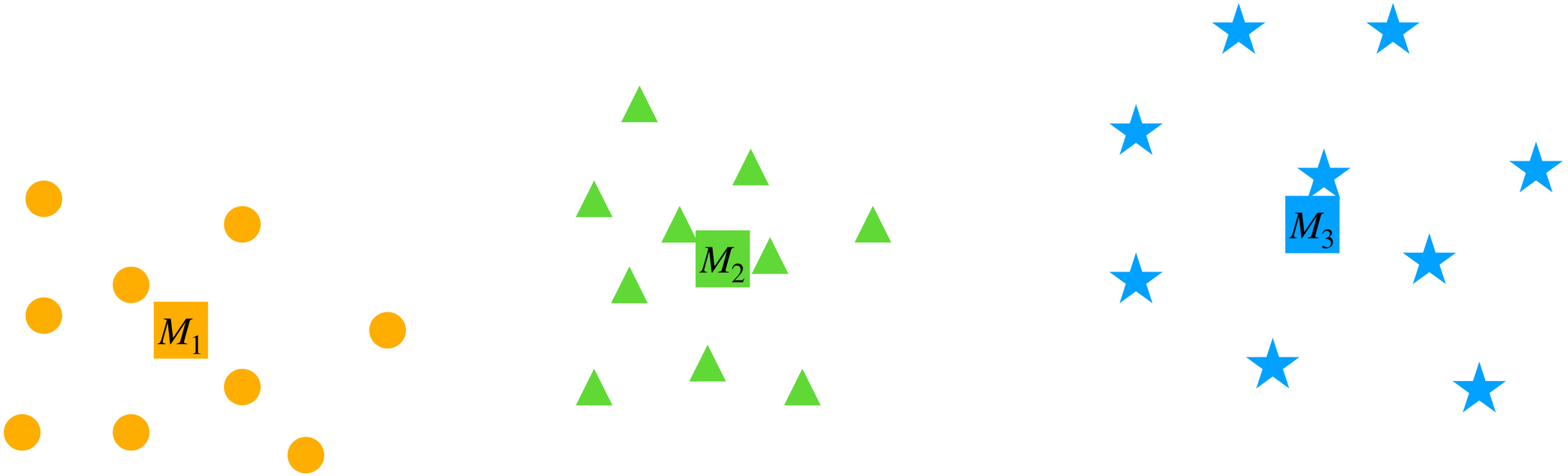
k-Means

Example



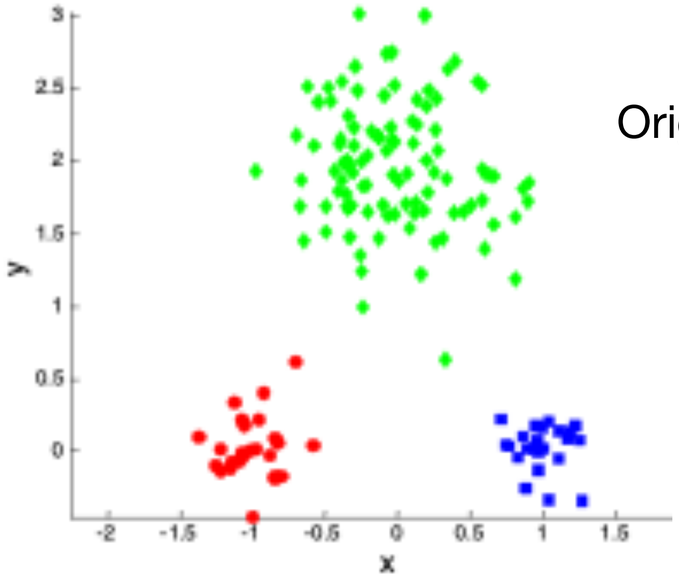
5. Repeat steps 3-4 until convergence.

k-Means

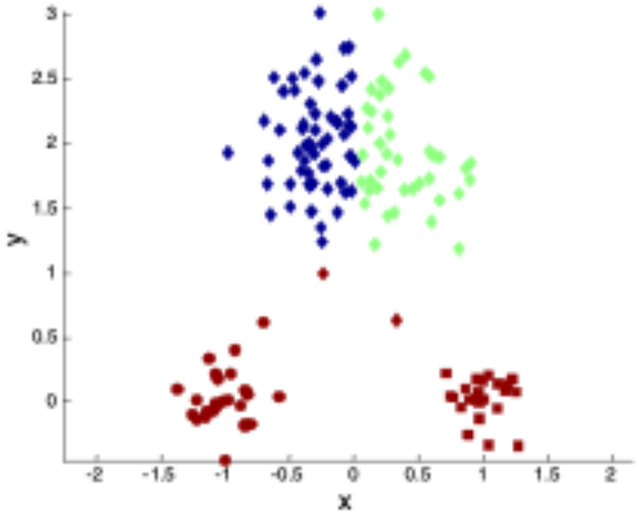
Advantages and Disadvantages

- Advantages:
  - Simple to understand.
  - Easy to compute.
- Disadvantages:
  - k must be manually specified.
  - Very sensitive to initial cluster centres.
  - Outliers can use up clusters.

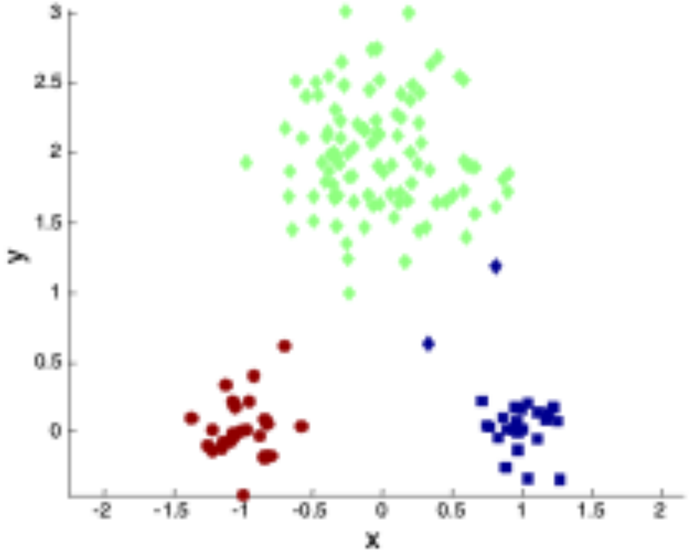
Two Different k-Means Clusterings



Original Data



Sub-optimal Clustering



Optimal Clustering

k-Means

Possible Improvements

- Run algorithm multiple times and choose the best result (the one with smallest total squared distance between data objects and closest cluster centre).

- Use improved versions of k-Means, such as k**-Means++** (chooses successive initial cluster centres with probability that is proportional to the square of the distance from the closest cluster centre).

k-Means Evaluation in Weka

```
=== Run information ===

Scheme:       weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core
Relation:     iris
Instances:    150
Attributes:   5
              sepallength
              sepalwidth
              petallength
              petalwidth
Ignored:
              class
Test mode:    Classes to clusters evaluation on training data

=== Clustering model (full training set) ===


kMeans
======

Number of iterations: 6
Within cluster sum of squared errors: 6.998114004826762

Initial starting points (random):

Cluster 0: 6.1,2.9,4.7,1.4
Cluster 1: 6.2,2.9,4.3,1.3
Cluster 2: 6.9,3.1,5.1,2.3

Missing values globally replaced with mean/mode
```

k-Means Evaluation in Weka

```
                           Cluster#
Attribute      Full Data          0          1          2
                 (150.0)     (61.0)     (50.0)     (39.0)
=============================================================
sepallength       5.8433     5.8885      5.006     6.8462
sepalwidth         3.054     2.7377      3.418     3.0821
petallength       3.7587     4.3967      1.464     5.7026
petalwidth        1.1987      1.418      0.244     2.0795



Time taken to build model (full training data) : 0.01 seconds

=== Model and evaluation on training set ===

Clustered Instances

0       61 ( 41%)
1       50 ( 33%)
2       39 ( 26%)


Class attribute: class
Classes to Clusters:

  0  1  2  <-- assigned to cluster
  0 50  0 | Iris-setosa
 47  0  3 | Iris-versicolor
 14  0 36 | Iris-virginica

Cluster 0 <-- Iris-versicolor
Cluster 1 <-- Iris-setosa
Cluster 2 <-- Iris-virginica

Incorrectly clustered instances :       17.0     11.3333 %
```

k-Means

- The quality of our clustering can be measured by the within-cluster variation:

$$E = \sum_{i=1}^{k} \sum_{p \in C_i} dist(\mathbf{p}, \mathbf{c_i})^2$$

- where E is the sum of the squared error for all objects in the data set, $\mathbf{p}$ is a given object, and $\mathbf{c_i}$ is the centroid of cluster C_i.

- Problem with k-Means: sensitivity to outliers.
  - When outliers are assigned to a cluster, they can dramatically alter the mean value of the cluster. Problem exacerbated due to use of **squared** error!

k-Medoids

- Consider seven objects in 1D space: 4,5,6,14,15,17,42
- What would be a good clustering? Maybe {4,5,6}, {14,15,17}, with 42 excluded (because it is an outlier)?
- Apply k-Means using k=2, the partitioning ({4,5,6},{14,15,17,42}) has the within-cluster variation:

$$(4-5)^2+(5-5)^2+(6-5)^2+(14-22)^2+(15-22)^2+(17-22)^2+(42-22)^2$$
$$=540$$

as the mean of {4,5,6} is 5, and the mean of {14,15,17,42} is 22.

Han, J., Pei, J. and Kamber, M., 2012. *Data Mining: Concepts and Techniques*. Elsevier.

k-Medoids

- Now let's try ({4,5,6,14},{15,17,42})
- $$(4-7.25)^2+(5-7.25)^2+(6-7.25)^2+(14-7.25)^2+(15-24.67)^2+(17-24.67)^2+(42-24.67)^2 = 515.4167$$

  as the mean of {4,5,6,14} is 7.25, and the mean of {15,17,42} is 24.67.
- As 515.4167<540, k-Means would assign 14 to the first cluster, due to the outlier (42).
- The centre of {15,17,42} is 24.67, which is much higher than all member of the cluster (except for the outlier).

Han, J., Pei, J. and Kamber, M., 2012. *Data Mining: Concepts and Techniques*. Elsevier.

k-Medoids

- How can we alter k-Means to prevent this problem?
- Instead of taking the mean value of the cluster, we can instead take one representative object per cluster.
- The remaining objects are assigned to the cluster where the representative object is the most similar.

k-Medoids

- Objective: Minimise the sum of dissimilarities between each object $\mathbf{p}$ and its representative object (most similar object), i.e.:

$$E = \sum_{i=1}^{k} \sum_{p \in C_i} dist(\mathbf{p}, \mathbf{o_i})$$

- E is the sum of the absolute error for all objects ($\mathbf{p}$). $\mathbf{o_i}$ is the representative object of cluster C_i.

k-Medoids

- Example k-Medoids algorithm: the Partitioning Around Medoids (Algorithm).

1. Arbitrarily choose the initial representative objects.
2. Replace the representative object with a non representative object, and see how this impacts clustering quality.
3. Step 2 continues until the quality of the resulting clustering cannot be improved.

Kaufman, L. and Rousseeuw, P.J., 2009. Finding groups in data: an introduction to cluster analysis (Vol. 344). John Wiley & Sons.

k-Medoids

- Let $\mathbf{o_1}, \ldots, \mathbf{o_k}$ be the set of current representative objects (medoids).
- Let $\mathbf{o_{random}}$ be a current nonrepresentative object, and we wish to determine whether $\mathbf{o_{random}}$ would be a good replacement for a current medoid $\mathbf{o_j}$ $(1 \leq j \leq k)$.
- Calculate the distance from every object $\mathbf{p}$ to the closest object in the set $\mathbf{o_1}, \ldots, \mathbf{o_{j-1}}, \mathbf{o_{random}}, \mathbf{o_{j+1}}, \ldots, \mathbf{o_k}$.

k-Medoids

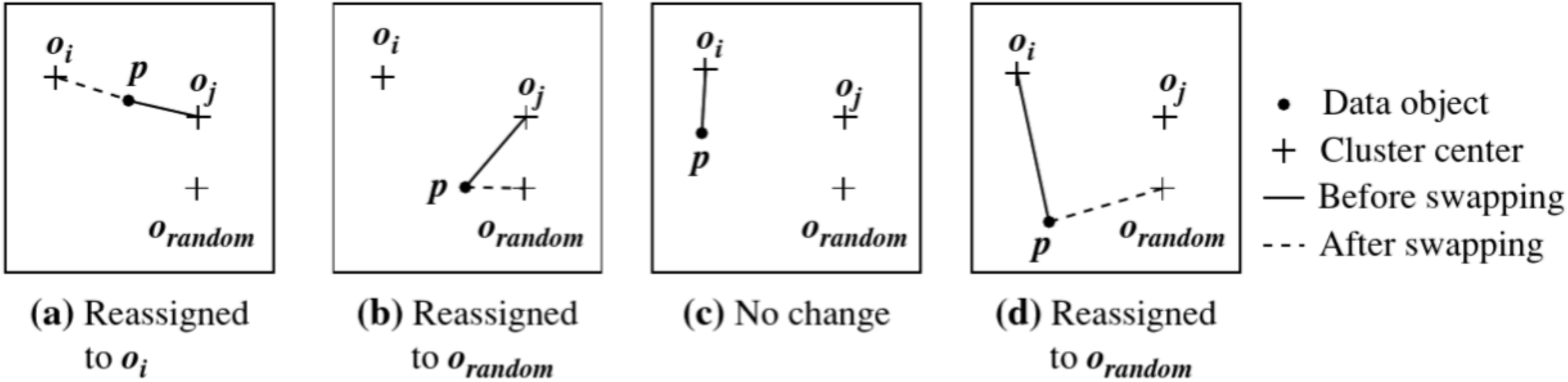


Figure from: Han, J., Pei, J. and Kamber, M., 2012. *Data Mining: Concepts and Techniques*. Elsevier.

k-Medoids

- Each time a reassignment occurs, the sum of dissimilarities between each object p and its representative object changes. Recall:

$$E = \sum_{i=1}^{k} \sum_{p \in C_i} dist(\mathbf{p}, \mathbf{o_i})$$

- The cost function calculates the difference in the absolute-error value if an existing representative object is replaced by a non-representative object.

- If the absolute error E is reduced, $\mathbf{o_j}$ is swapped with $\mathbf{o_{random}}$, otherwise, there is no change.

k-Medoids vs k-Means

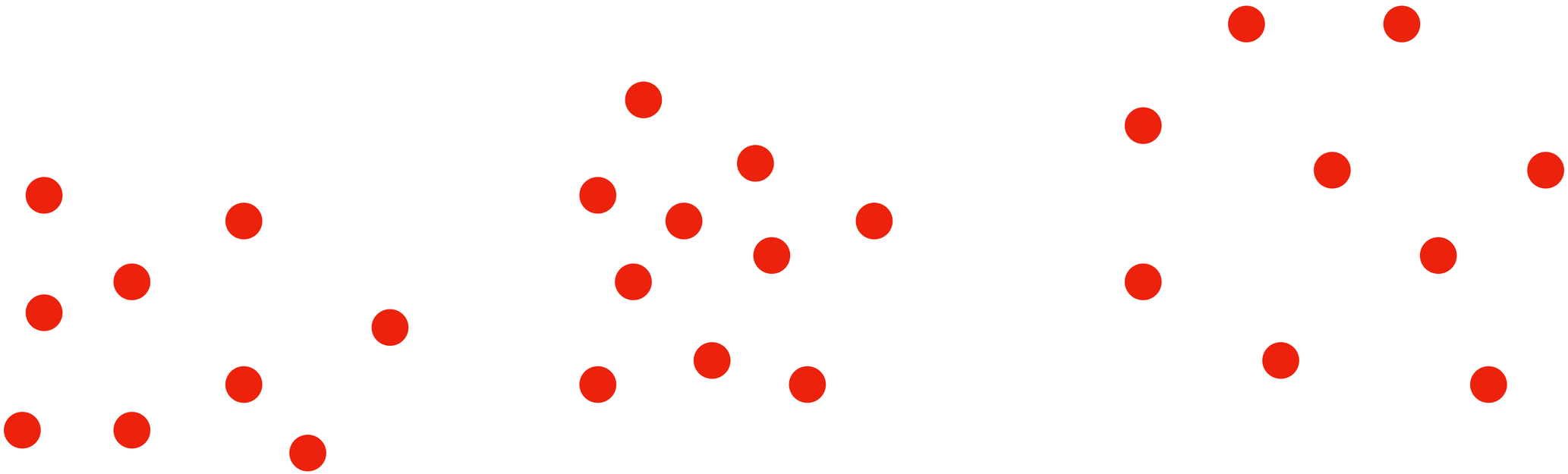
- k-Medoids is more robust than k-Means to noise and outliers.
  - A medoid is less influenced by noise and outliers.
- However, k-Medoids become very costly for large values of n and k.
- Disadvantage of both: k must be specified!

k-Centers

- Want to minimise diameter of each cluster.
  - Pick some object from the data as the first centre. Repeat:
    - For each data object, compute its distance d_{min} from its closest centre
    - Find the data object that maximises d_{min}
    - Add this object to the set of centres
  - Until k centres are picked
- An example is *Farthest First Clustering*.

Farthest First Clustering, k=3

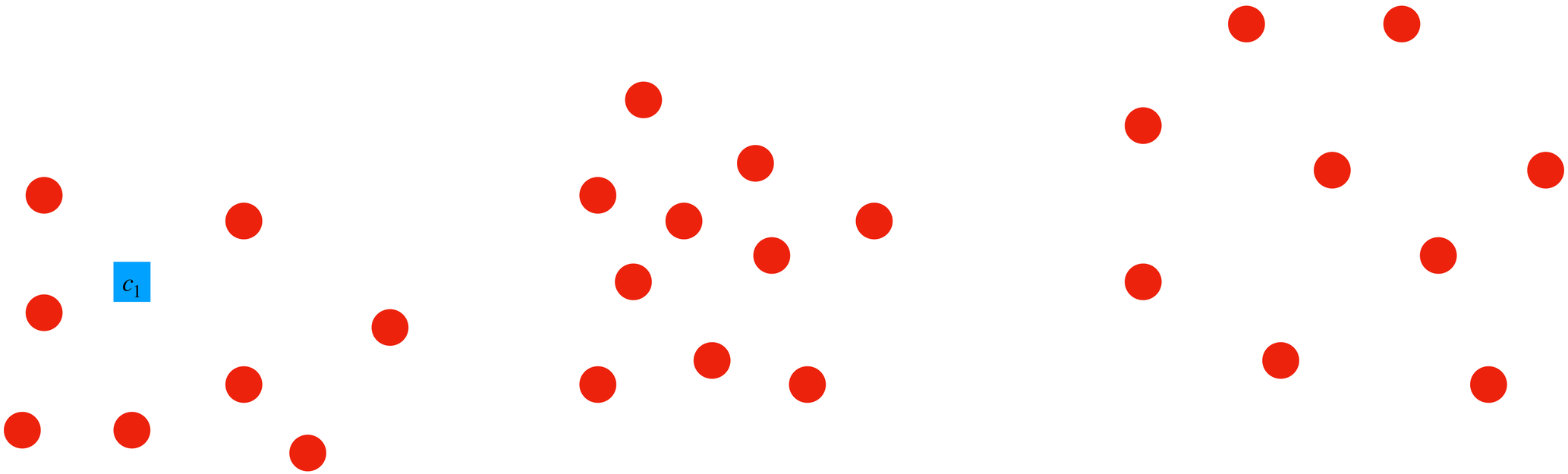
(Also called Furthest Point Clustering)



Initial data.

Farthest First Clustering, k=3

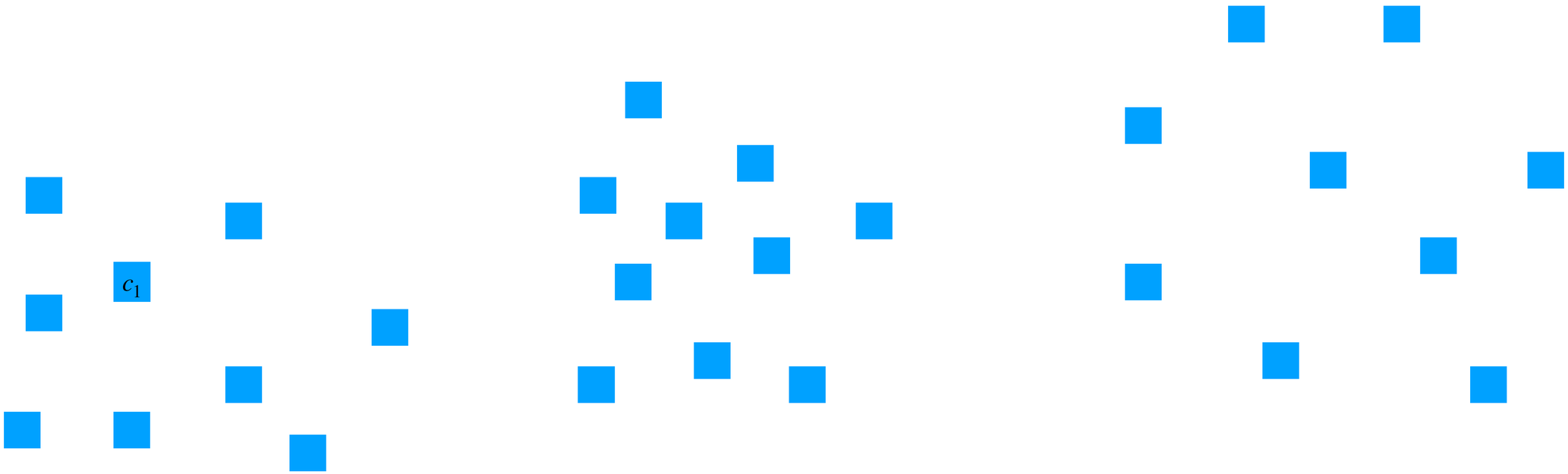
(Also called Furthest Point Clustering)



Step 1: Choose an arbitrary centre, c_1.

Farthest First Clustering, k=3

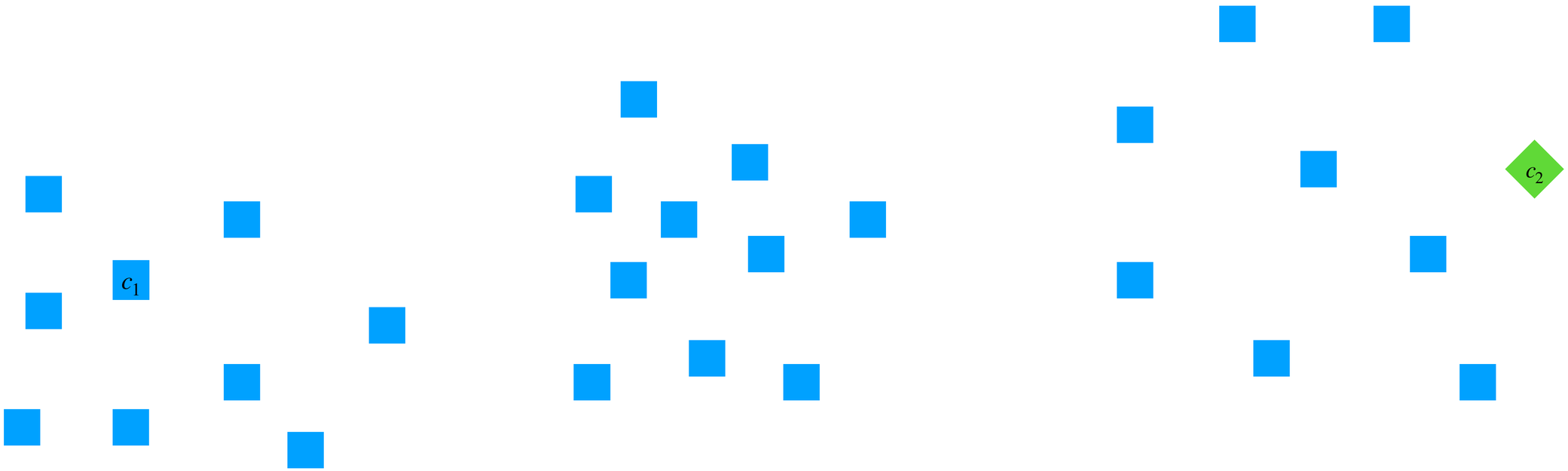
(Also called Furthest Point Clustering)



Step 2: Assign objects to their closest centre.

Farthest First Clustering, k=3

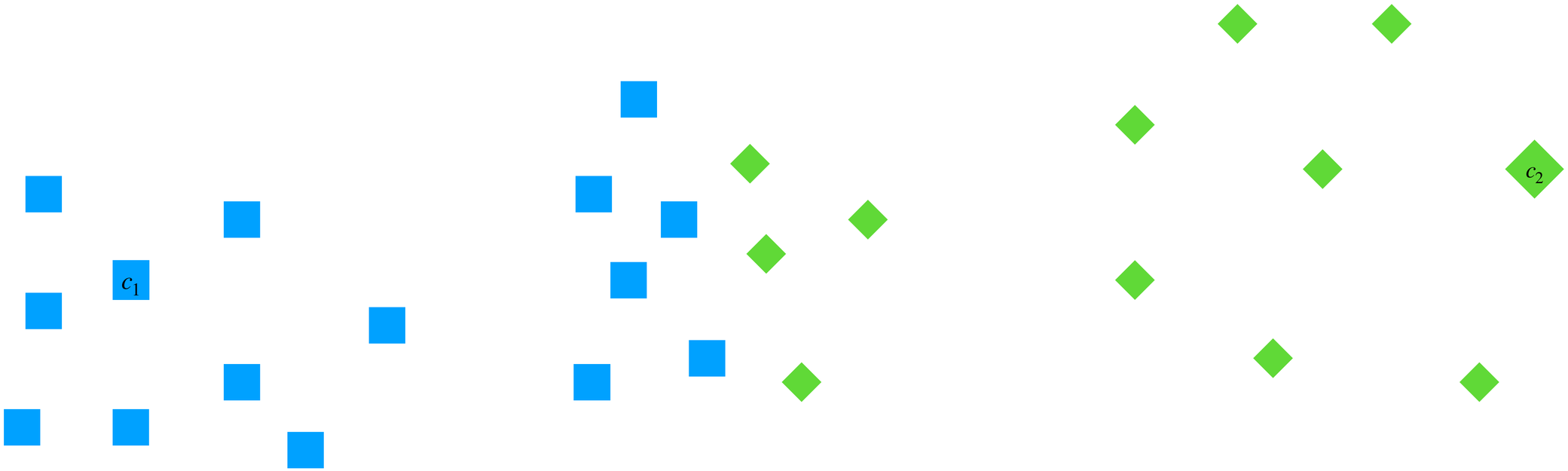
(Also called Furthest Point Clustering)



Step 3: Select the next centre, c_2, to be the one farthest from its closest centre.

Farthest First Clustering, k=3

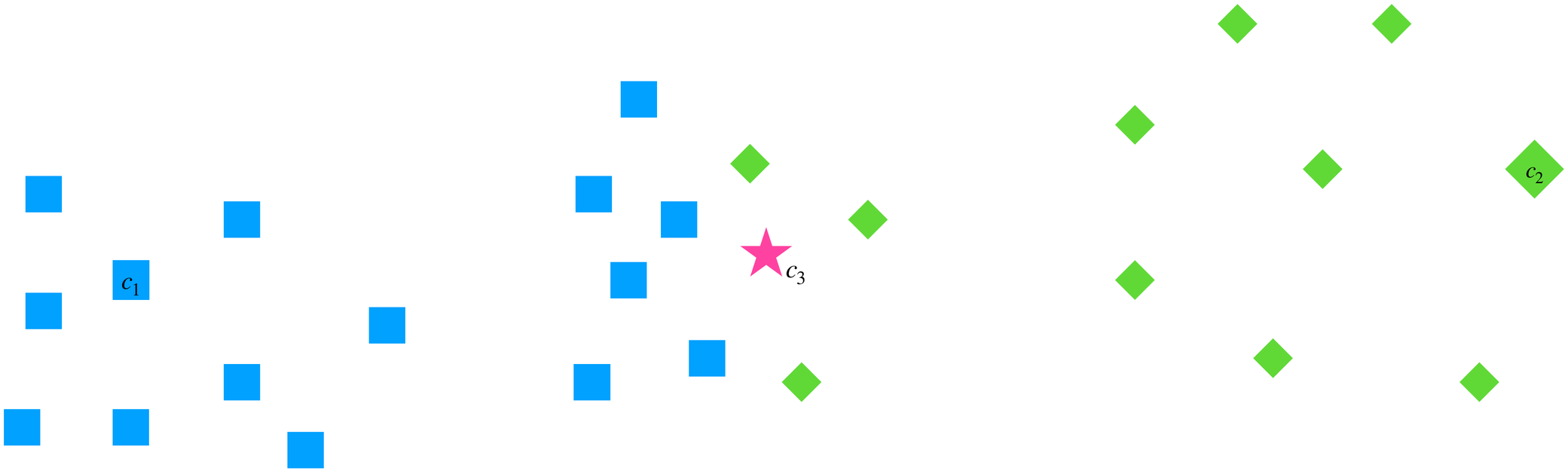
(Also called Furthest Point Clustering)



Step 4: Assign objects to their closest centre.

Farthest First Clustering, k=3

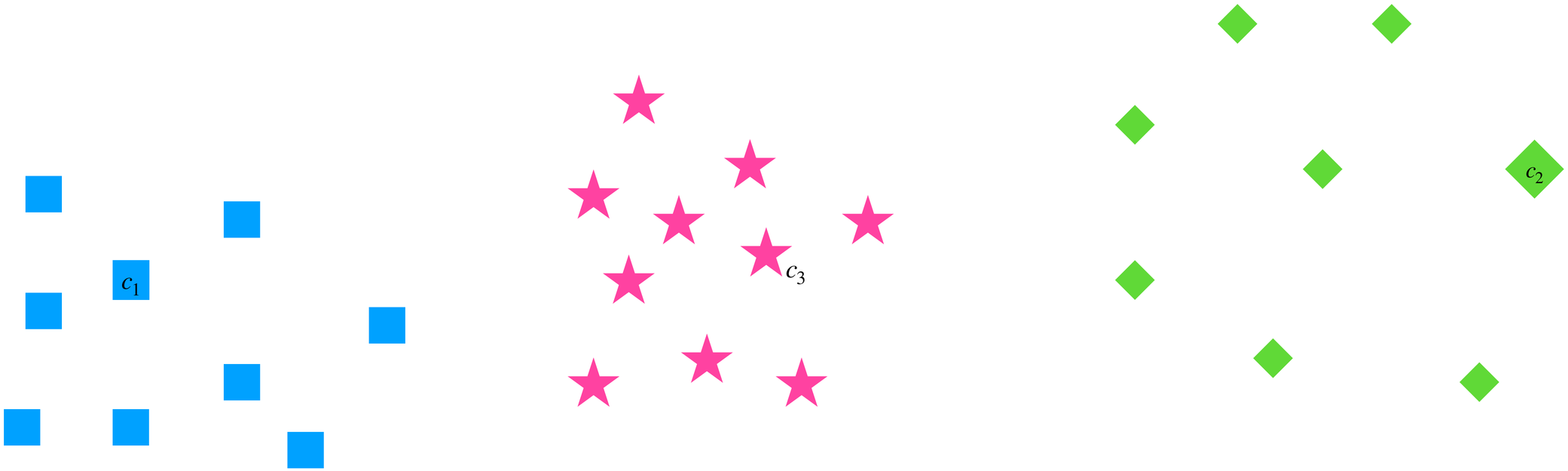
(Also called Furthest Point Clustering)



Step 5: Select the next centre, c_3, to be the one farthest from its closest centre.

Farthest First Clustering, k=3

(Also called Furthest Point Clustering)



Step 6: Assign objects to their closest centre.

Cluster Evaluation

- Harder to evaluate quality of a clustering
  - Can measure how well a method does based on its own criteria
  - But each method has different objectives
- Can use a (withheld) class attribute to define a “true” clustering
  - Assumes that the points cluster well based on class
  - Good for evaluating new methods against known data
  - Not good for evaluating known methods against new data!
  - Study the **confusion matrix** comparing class to clusters

Cluster Evaluation

- Will use the famous *iris* data set to evaluate clustering
  - Contains 50 samples from each of three types of iris (flower)
  - Measures sepal length, sepal width, petal length, petal width
  - In the Weka default data sets as *iris.arff*

Farthest First Clustering in Weka

```
Scheme:weka.clusterers.FarthestFirst -N 3 -S 1
Relation:     iris
Instances:    150
Attributes:   5
      sepallength
              sepalwidth
              petallength
              petalwidth
Ignored:      class
Test mode:Classes to clusters evaluation on training data

=== Model and evaluation on training set ===
Cluster centroids:

Cluster 0
  7.7 3.0 6.1 2.3
Cluster 1
  4.3 3.0 1.1 0.1
Cluster 2
  4.9 2.5 4.5 1.7
Time taken to build model (full training data) : 0 seconds
```

We set k = 3 (using domain knowledge)

Farthest First Clustering in Weka

```
=== Model and evaluation on training set ===
Clustered Instances
0       41 ( 27%)
1       50 ( 33%)
2       59 ( 39%)
Class attribute: class
Classes to Clusters:
  0  1  2  <-- assigned to cluster
  0 50  0 | Iris-setosa
  6  0 44 | Iris-versicolor
 35  0 15 | Iris-virginica

Cluster 0 <-- Iris-virginica
Cluster 1 <-- Iris-setosa
Cluster 2 <-- Iris-versicolor

Incorrectly clustered instances :  21.0  14      %
```

Acknowledgments

- Greg Watson [Warwick, CS430/CS910]
- Han, J., Pei, J. and Kamber, M., 2012. *Data Mining: Concepts and Techniques*. Elsevier.
- Graham Cormode [Warwick, CS910]
- Florin Ciucu [Warwick, CS430/CS910]